

MOSFET Regulator



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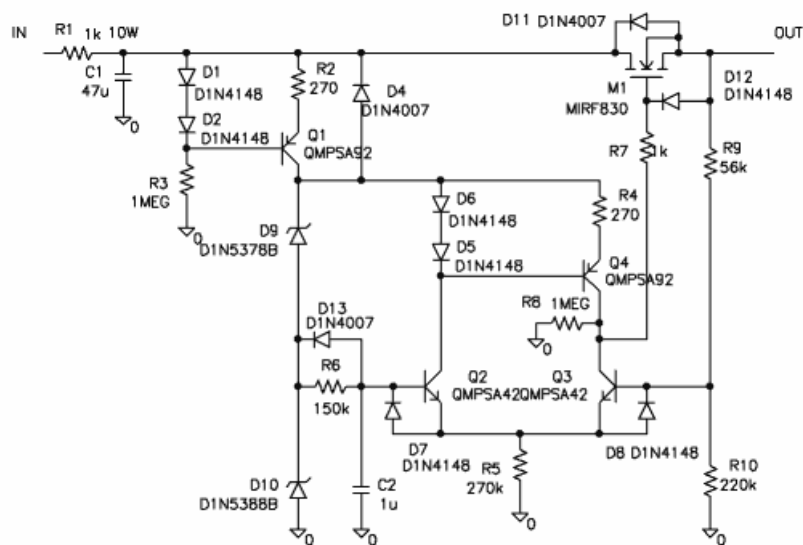
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The DDS-2250V uses a MOSFET regulator to provide all the high voltage supplies with the exception of the anode and screens of the output valves.

Input can be in the range 325-500V (nominally 430V), output is fixed at 250V. In the DDS-2250V, about 25mA is being drawn from the regulator, so only a small heatsink is required on the IRF830.

Heres the circuit diagram:



DDS-2250V MOSFET HV Regulator - Design Copyright ©1997 Duncan Munro

The circuit was drawn with Pspice from Orcad.

[Click here](#) if you want to download a clearer version of the circuit in Adobe Acrobat (pdf) format.

Circuit Description

R1 and C1 clean up the input, removing the high amount of ripple which is present in the supply.

Q1 is a constant current source which feeds the zener diode stack, 300V appears at the anode of D9 and 200V appears at the anode of D10. R6 and C2 clean up any noise from the zeners and feed into Q2, the first part of the differential amplifier. Q4 is another constant current source which acts as the load for the second part of the differential amplifier, Q3.

R7 interacts with the MOSFET input capacitance to limit the high speed response of the feedback circuit. The voltage divider of R9 and R10 is for feedback sensing, and aims to put about 200V on the base of Q3.

The voltage rating of the transistors used is 300V (except the IRF830 which is 500V). All the extra diodes are there to prevent things blowing up when switching off.

Is it over the top?

The idea started in 1989 when I liked the functionality of three terminal regulators, such as the 7805 and 7812, but wanted to apply this "black box" approach to higher voltage applications. At that time, the project was a regulator for a colour monitor chassis needing a feed of 115V DC at 0.4A. My older design measured 2mV ripple on the output, and this has been improved for the guitar amplifier regulator you see above to get down to microvolts.

You could arguably use a couple of zeners to give 250V, and just feed into the gate of the MOSFET. This will work, however output impedance will be higher, so will noise. More filtering will be required downstream.

Most amplifiers use a choke to smooth things out - in fact, this is what I was going to do until I worked out that the circuit above gave a cleaner output. The MOSFET regulator costs slightly less than a choke, however you need to put it on a PCB of some sort which may tip your views on the convenience side of things.....